

Vibe-Pythonpass: Production & Operational Monitoring Process

- **Development & CI/CD Pipeline**

- Code repository: Private GitHub with branch protection rules
- Build: Automated Docker image builds on commit to main
- Testing: Automated unit tests (pytest), integration tests, security scanning (bandit)
- Deployment: Blue-green deployment strategy, automated rollback on failure
- Release cadence: Two-week sprints, hotfixes as needed

- **Production Monitoring**

- Application Performance Monitoring (APM): DataDog with 15-second granularity
- Log Aggregation: ELK stack with 30-day retention
- Metrics Collection: Prometheus + Grafana dashboards
- Uptime Monitoring: Synthetic monitoring every 5 minutes
- SLA Target: 99.95% uptime (max 22 minutes downtime/month)

- **Incident Response**

- On-call rotation: 24/7 coverage, response target: 15 minutes for P1
- Incident classification: P1 (service down), P2 (partial outage), P3 (degradation)
- Post-incident reviews: Within 24 hours for P1 incidents
- RCA documentation required for all incidents

- **Database Maintenance**

- Automated backups: Hourly snapshots, 90-day retention

- Backup verification: Daily restore tests from backups
- Integrity checks: Daily VACUUM and ANALYZE operations
- Storage optimization: Archive inactive accounts after 2 years of inactivity
- **Security Monitoring**
 - Failed login attempts: Alert if >10 per minute
 - Unusual API access patterns: ML-based anomaly detection
 - Dependency vulnerability scanning: Daily (Snyk, Dependabot)
 - WAF (Web Application Firewall): Rate limiting, DDoS protection